

Wh-doublets in Cantonese, and their typological implications

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Table of contents

- ① Introduction
- ② Background: *Wh*-doublets
- ③ Bare vs. non-bare
- ④ Plurality
- ⑤ Derivation
- ⑥ Evidential indefinites
- ⑦ Conclusions

Introduction

- As with many languages, Cantonese *wh*-expressions can be reduplicated.

① **Bare** *wh*-doublets: [XP *wh-wh*] (category-preserving)

- (1) 阿明話 [佢嚟自助餐食咗 乜野乜野] 。 SG/PL

Aaming waa [keoi hai zizocaan sik-zo matje matje]

Aaming say he at buffet eat-PFV what what

(lit.): 'Aaming said he ate **what what** at the buffet.'

- But what is less discussed is that *wh*-reduplication can take multiple forms.

Introduction

② Non-bare, nominal *wh*-doublets: [Nominal *wh*+ α *wh*+ α]

- (2) 阿明話 [佢喺自助餐食咗 乜野蛋糕乜野蛋糕] 。 PL only

Aaming waa [keoi hai zizocaan sik-zo matje daangou matje daangou]

Aaming say he at buffet eat-PFV what cake what cake

(lit.): 'Aaming said he ate **what cake what cake** at the buffet.'

③ Non-bare, clausal *wh*-doublets: [Clause *wh*+ α *wh*+ α]

- (3) 阿明話 [佢喺自助餐 食咗乜野食咗乜野] 。 PL only

Aaming waa [keoi hai zizocaan sik-zo matje sik-zo matje].

Aaming say he at buffet eat-PFV what eat-PFV what

(lit.): 'Aaming said he **ate what ate what** at the buffet.'

More naturally occurring examples

• Bare *wh*-doublets

- (4) a. 邊間機構捐錢唔捐錢，捐幾多都要比較。

邊個邊個 有無出post哀悼又要批評。

who who

- b. 就算我有自己嘅習慣用法，我都淨係會話，
我因為某某原因，所以會 點樣點樣 寫。

how how

- c. 最近身邊好多朋友吹爆「蒸爐」，
話佢 點解點解 好、 邊度邊度 靚。

why why, where where

• Non-bare *wh*-doublets

- (5) a. 我夠識講呢個人 邊度唔好邊度唔好。

what bad what bad

- b. 我經常教fd 點樣做點樣做，但到自己又做唔出。

how do how do

- c. 越攞越唔掂，個人又成日好蚊佢 點解噉點解噉。

why so why so

Introduction

Key properties of Cantonese *wh*-doublets:

① **Productivity** of both types

② ***Wh*-indefinites:**

All of them involve an existential interpretation of *wh*-expressions.

③ **Morpho-syntactic differences** in bare and non-bare *wh*-doublets

④ **Plurality** in non-bare *wh*-doublets, but not in the bare counterparts

⑤ **Evidentiality**

They all convey an evidential implicature, different from the more studied existential *wh*-singlet cases (Li 1992; Lin 1998).

→ Cantonese informs us about the typology of *wh*-doublets and indefinites.

Table of contents

- ① Introduction
- ② Background: *Wh*-doublets
- ③ Bare vs. non-bare
- ④ Plurality
- ⑤ Derivation
- ⑥ Evidential indefinites
- ⑦ Conclusions

Wh-doublets

(Bare) *wh*-singlets

- ① interrogative or non-interrogative (or both)
- ② number-neutral

(6) *Wh*-singlets

- a. Who has come? (A: John./ John and Mary.) interrogatives only
- b. *Mandarin*
 you shei lai-le (Ching-yuan 1990)
 seems who come-PFV
 (i) ‘Who has come?’ (A: John./ John and Mary.) interrogatives
 (ii) ‘There is somebody coming.’ indefinites

Wh-doublets across languages

Wh-doublets

- ① a similar interrogative vs. indefinite split
- ② but not always number-neutral

(7) *Wh*-doublets in Korean (Chung 1999, p.282-283)

- a. *wh*-interrogatives → plural

nwukwu-nwukwu o-ess-ni?
 who-who come-PST-Q
 ‘Who_{plural} came?’

- b. *wh*-indefinites → singular

yeysnal-ey sokumcangswu-ka eti-eti -lul ka-ess-nuntey
 long;time;ago salt;seller-NOM where-where-ACC go-PST-when

Singular: ‘A long time ago, a salt seller went to **a place**.’

Plural: %‘A long time ago, a salt seller went to **some (several) places**.’

Wh-doublets across languages

- More commonly, *wh*-doublets are either

- 1 plural interrogatives, or
- 2 number-neutral indefinites

(8) Yaeyaman, a southern Ryukyuan language (Davis 2016, p.637)

- a. sinu=ya taa=du kitta-riyaa.
 yesterday=TOP who=FOC came-Q
 ‘**Who**_{SG} came yesterday?’
- b. sinu=ya taa+taa=du kitta-riyaa.
 yesterday=TOP who+who=FOC came-Q
 ‘**Who**_{PL} came yesterday?’

Wh-doublets across languages

- Indefinite usages of *wh*-doublets are often described as allowing for a free choice reading, without mentioning plurality.

(389) (a) **Moksha Mordva** (Finno-Ugrian; Feoktistov 1966: 207)

<i>kije</i>	‘who?’	<i>kije-kije</i>	‘somebody, anybody’
<i>meze</i>	‘what?’	<i>meze-meze</i>	‘something, anything’
<i>kodama</i>	‘which?’	<i>kodama-kodama</i>	‘any’

(b) **Ila** (Bantu)

<i>oni</i>	‘who?’	<i>oni-oni</i>	‘whoever, anyone’
<i>ukwi</i>	‘where?’	<i>ukwi-ukwi</i>	‘anywhere’

(c) **Vietnamese** (Coyaud and Aït Hamou 1972)

<i>ai</i>	‘who?’	<i>ai ai</i>	‘anybody, everybody’
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(d) **Latin**

<i>quis</i>	‘who?’	<i>quis-quis</i>	‘anybody, whoever’
<i>ubi</i>	‘where?’	<i>ubi-ubi</i>	‘anywhere, wherever’

(e) **Bugis (Austronesian)**; Sirk 1979: 116)

<i>niga</i>	‘who?’	<i>niga-niga</i>	‘anyone’
<i>aga</i>	‘what?’	<i>aga-ga</i>	‘anything’

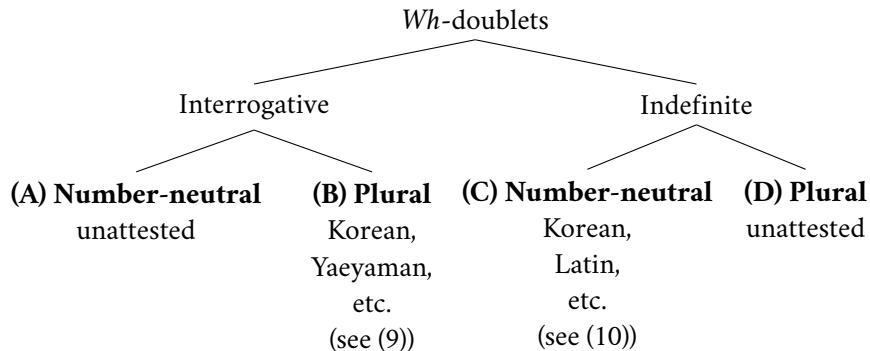
(f) **Ainu** (Refsing 1986: 104)

<i>nen</i>	‘who?’	<i>nen-nen</i>	‘whoever, nobody’
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(Haspelmath 1997)

Wh-doublets across languages

Types of *wh*-doublets across languages



Wh-doublets across languages

- Type (B): interrogative, plural *wh*-doublets

(9)	a.	Korean	(Kim 1999; Chung 1999)
	b.	Japanese	(Kudo:2021)
	c.	Okinawan (Kin)	(Tamamoto:2021)
	d.	Yaeyaman	(Davis 2016)
	e.	Mongolia (Khalkha)	(Jang&Lim:2024)
	f.	Tibetan-Burmese	(Rigzinwangmo:1987; Zhang:2016a)
	g.	Guinean Maninka	(Vydrin&Diane:2016)
	h.	Tagalog	(Hong:2016)
	i.	Turkish	(Lim&Jang:2024)
	j.	Korku	(Nagaraja:1999)

Wh-doublets across languages

- Type (C): indefinite, number-neutral *wh*-doublets

- | | | | |
|------|----|---------------|-------------------|
| (10) | a. | Korean | (Chung 1999) |
| | b. | Latin | (Haspelmath 1997) |
| | c. | Ila | (Haspelmath 1997) |
| | d. | Hindi/Urdu | (Haspelmath 1997) |
| | e. | Mandarin | (Xu 2010) |
| | f. | Japanese | (Sudo 2008, 2013) |
| | g. | Ainu | (Refsing 1986) |
| | h. | Khasi | (Moravcsik 1978) |
| | i. | Kristang | (Baxter 1988) |
| | j. | Moksha Mordva | (Feoktistov 1966) |

Wh-doublets across languages

- It is tempting to draw the following generalization:

- (11) a. When used **interrogatively**, *wh*-doublets must be construed as **plural**.
- b. When used as **indefinites**, *wh*-doublets must be construed as **number-neutral**.
- We suggest that Cantonese non-bare *wh*-doublets instantiate the unattested Type (D).
 - they are indefinites with obligatory plural reading.
 - cf. bare *wh*-doublets are number-neutral, hence Type (C).

Table of contents

- 1 Introduction
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- 4 Plurality
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- [illegible]

Table 1: The morpho-syntactic differences of bare and non-bare *wh*-doublets

#1 Heads vs. phrases

- **Bare *wh*-doublets** target any *wh*-heads.

(12) 又話 邊個邊個 賣樓又蝕咗幾多錢 source

jau waa bingo bingo maai lau jau sit-zo geido cin
 again say who who sell house again lose-PFV how.much money
 (lit.): ‘(They) again said **who who** sold flats and lost how much money.’

(13) 阿明話 [佢喺自助餐 點樣點樣 食晒啲生蠔] 㗎。

Aaming waa [keoi hai zizocaan dimjoeng dimjoeng sik-saai di
 Aaming say he at buffet how how eat-all CL
 saanghou] wo5
 oyster SFP

(lit.): ‘Aaming said he ate all the oysters at the buffet in **how how** (ways).’

#1 Heads vs. phrases

- **Non-bare *wh*-doublets** apply to phrases containing a *wh*-expression.

① *Nominal cases*: Adjective/Relative clause + NP

- (14) 阿明岩岩講起佢要上 幾時嘅堂幾時嘅堂 。 PL
 Aaming aam-aam gong-hei keoi jiu soeng geisi-ge tong geisi-ge tong
 Aaming just.now mention he need attend when-MOD class
 (lit.): ‘Aaming just talked about **classes at what time classes at what time** he has to attend.’
- (15) 老闆喺度解釋點解想請 黎自邊度嘅人黎自邊度嘅人 。 PL
 loubaan haidou gaaisik dimgaai soeng ceng lai-zi bindou-ge jan lai-zi bindouge jan
 boss PROG explain why want hire come-from where-MOD person
 (lit.): ‘The boss is explaining why (he) wants to hire **people from where people from where**.’

#1 Heads vs. phrases

② Clausal cases: beyond VPs

- (16) 阿明成日話 邊個食素邊個食素 , 搞到有晒營養。 PL

Aaming sengjat waa bingo sik sou bingo sik sou , gaau-dou mou-saai

Aaming always say who eat vegetarian, make-RES not.have-all

jingjoeng.

nutrition

(lit.): 'Aaming always says **who is vegetarian who is vegetarian**, so they take in no nutrition at all.'

- (17) 阿明不停話佢今年 去咗邊度黎去咗邊度黎 。 PL

Aaming batting waa keoi gam-nin heoi-zo bindou lai4 heoi-zo bindou lai4 .

Aaming repeatedly say he this-year go-PFV where SFP

(lit.): 'Aaming kept saying **where he went where he went** this year.'

#2 Constituency

- The boundary of reduplication differs in the two types of *wh*-doublets.
- **Bare *wh*-doublets** need not obey syntactic constituency.

(18) 隨隨便便走上嚟就話 邊本邊本 書有「煽動意圖」。PL
 ceoiceoibinbin zau-soeng lai zau waa bin-bun bin-bun syu
 casually walk-up come then say which CL which CL book
 jau “sindung jitou”
 have seditious intention
 (lit.): ‘They casually came up and said **which which** books have
 “seditious intention”.

→ The determiner *bin* and classifier *bun* do not form a constituent.

#2 Constituency

- **Non bare *wh*-doublets** observe constituency.

(19) *Subject-verb string

*阿明成日都同我講 邊個去邊個去 日本。

*Aaming sengjat dou tung ngo gong bingo heoi bingo heoi Jatbun.

Aaming always also with me say who go who go Japan
 (lit.): 'Aaming always tells me **who went who went** to Japan.'

(20) *Adverb-verb string

*阿明成日叫人 點樣整點樣整 芝士。

*Aaming sengjat giu jan dimjoeng zing dimjoeng zing zisi-beng.

Aaming always ask person how make how make cheese-cake
 (lit.): 'Aaming always asks people **how to make how to make** cheese cake.'

#3 Triplication

- **Bare *wh*-doublets**
- Triplication is possible for bare *wh*-doublets when the *wh*-expression is monosyllabic.
- ‘What’ in Cantonese has two forms: *matje* vs. *mat*

(21) *Wh*-triplets are sensitive to the form of *wh*-expressions

阿明話佢用咗 { 乜乜乜 } 黎整蛋糕。
 *乜野乜野乜野

Aaming waa keoi jung-zo mat mat mat / *matje matje matje lai zing
 Aaming say he use-PFV what what what to make
 daangou.
 cake

(lit.): Aaming said he used **what what** to make a cake.’

#3 Triplication

- **Non-bare *wh*-doublets**
- Triplication is impossible for the non-bare *wh*-doublets, even the *wh*-expression is monosyllabic.

(22) No triplication

*阿明話佢 食咗乜食咗乜食咗乜 先訓覺。

Aaming waa keoi sik-zo mat sik-zo mat sik-zo mat sin fangaau
 Aaming say he eat-PFV what then sleep

(lit.): 'Aaming said he will sleep after **eating what eating what eating what.**'

Interim summary

- Bare and non-bare *wh*-doublets are morpho-syntactically different.

<i>Wh</i>-doublets	Bare	Non-bare
1. Reduplicated element	heads	phrases
2. Constituency	not always obeyed	obeyed
3. Triplication	✓ for monosyllabic <i>wh</i>	✗

Table 2: The morpho-syntactic differences of bare and non-bare *wh*-doublets

Table of contents

- 1 Introduction
- 2 Background: *Wh*-doublets
- 3 Bare vs. non-bare
- 4 Plurality**
- 5 Derivation
- 6 Evidential indefinites
- 7 Conclusions

Plurality

- Intuitively, non-bare *wh*-doublets convey a clearer sense of plurality, compared to bare *wh*-doublets.
- To differentiate the two types of *wh*-doublets in semantic number, it is important to establish **discourse contexts that strongly favor a singular reading**.
 - bare *wh*-doublets → compatible (if they are number-neutral)
 - non-bare *wh*-doublets → incompatible (if they are obligatorily plural)

Plurality

- Case 1: non-bare nominal *wh*-doublets

(23) 阿明話佢喺 邊間邊間 醫院出世㗎。 bare: sg

Aaming waa keoi hai bin-gaan bin-gaan jijyun ceotsai wo5.

Aaming say he at which-CL which-CL hospital born SFP

(lit.): Aaming said he was born at **which which** hospital.'

(24) # 阿明話佢喺 邊間醫院邊間醫院 出世㗎。 non-bare: #PL

#Aaming waa keoi hai bin-gaan jijyun bin-gaan jijyun ceotsai wo5.

Aaming say he at which CL hospital born SFP

(lit.): Aaming said he was born at **which hospital which hospital**.'

Plurality

- **Case 2: non-bare clausal *wh*-doublets, with *wh*-arguments**

(25) 阿明有提到嗰個同學姓乜野乜野。 bare: SG

Aaming jau taidou go-go tunghok sing matje matje

Aaming have mention that-CL classmate surname what what
(lit.): Aaming mentioned that classmate's surname is **what what**.'

(26) #阿明有提到嗰個同學 姓乜野姓乜野。 non-bare: #PL

#Aaming jau taidou go-go tunghok sing matje sing matje

Aaming have mention that-CL classmate surname what surname what
(lit.): Aaming mentioned that classmate **surnamed what surnamed what**.'

• Case 3: non-bare clausal *wh*-doublets, with *wh*-adjuncts

- (27) 專家都指出同一個原因，解釋層樓 點解點解 㗎。 bare: SG

zyungaa dou ziceot tung jat-go jyunjan, gaaisik cang lau
 expert all point.out same one-CL reason, explain CL building
 dimgaai dimgaai lam.

why why collapse

(lit.): Experts all pointed out the same reason, explaining **why why** the building collapsed.'

- (28) # 專家都指出同一個原因，解釋層樓 點解㗎點解㗎。 non-bare: #PL

#zyungaa dou ziceot tung jat-go jyunjan, gaaisik cang lau
 expert all point.out same one-CL reason, explain CL building
 dimgaai lam dimgaai lam .

why collapse why collapse

(lit.): Experts all pointed out the same reason, explaining the building **why collapsed why collapsed**.'

Interim summary

- So the desiderata:

<i>Wh</i>-doublets	Bare	Non-bare
1. Reduplicated element	heads	phrases
2. Constituency	not always obeyed	obeyed
3. Triplication	✓ for monosyllabic <i>wh</i>	✗
4. Number	number-neutral	obligatorily plural

Table 3: The differences of bare and non-bare *wh*-doublets

Table of contents

- 1 Introduction
- 2 Background: *Wh*-doublets
- 3 Bare vs. non-bare
- 4 Plurality
- 5 Derivation**
- 6 Evidential indefinites
- 7 Conclusions

Proposal

- We adopt a non-uniform approach to *wh*-doublets in Cantonese.
- They are derived via different reduplication morphemes.
 - Bare *wh*-doublets → phonological reduplication
 - Non-bare *wh*-doublets → syntactic reduplication

Two RED-morphemes	RED-affix	RED
Reduplication type	Phonological	Syntactic
Morphology	Affix	Free morpheme
Syntax (C-selection)	a <i>wh</i> -expression	a phrase containing a <i>wh</i>
At-issue meaning	An identity function	A plural operator

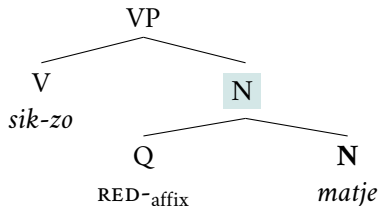
Table 4: The two proposed RED-morphemes

Derivation of bare *wh*-doublings

- **Bare *wh*-doublings** involve an affixal RED-morpheme, and reduplication occurs at PF.

- (29) 阿明話 [佢喺自助餐食咗 乜嘢乜嘢] 。 SG/PL
 Aaming waa [keoi hai zizocaan sik-zo matje matje]
 Aaming say he at buffet eat-PFV what what
 (lit.): 'Aaming said he ate **what what** at the buffet.'

- (30) Phonological reduplication in bare *wh*-doublings (Travis 2001, 2003)



Evidence #1

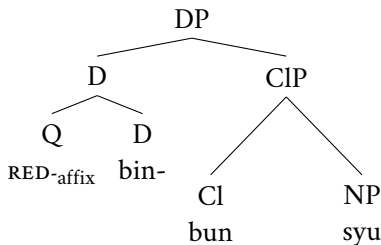
- Deriving the properties:

- (31)
- Heads only → RED is an affix.
 - Constituency violation → reduplicaiton occurs at PF.
 - Triplication → constrained by prosodic structures
 - Number neutrality → RED is an identity function

Evidence #2

- Remarks on constituency violation: how it is derived
- Recall that the Det-Cl string can be reduplicated, i.e., *bin-bun bin-bun*

(32) a. Syntax



b. PF: Affixation of *bin-* precedes reduplication

Step 1 - Affixation of *bin-* on *bun* → *bin-bun*

Step 2 - Reduplication of *bin-bun* → *bin-bun bin-bun*

Evidence #2

- A prediction is that if reduplication occurs before affixation, we expect to see the triplication of *bin-* (of *bingo* ‘who’).

(33) 我丫婆成日同丫姨講見到 邊邊邊個 。 source
 ngo aapo sengjat tung aaji gong gindou bin bin bin go .
 my grandma always with auntie say see who who who CL
 (lit.): ‘My grandma always tells auntie she saw **who who who**.’

- In sum, bare *wh*-doublets are constrained not by syntactic constituency, whose forms are sensitive to phonological properties.
- These observations follow from the phonological nature of the RED-morpheme.

Derivation of non-bare *wh*-doublets

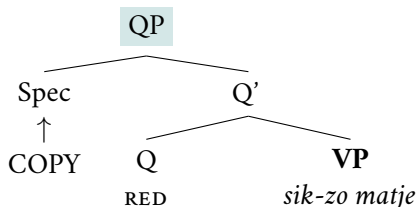
- **Non-bare *wh*-doublets** involve a free RED-morpheme, and reduplication occurs at syntax.

(34) 阿明話 [佢喺自助餐 食咗乜嘢食咗乜嘢] 。 PL only

Aaming waa [keoi hai zizocaan sik-zo matje sik-zo matje]

Aaming say he at buffet eat-PFV what eat-PFV what
(lit.): 'Aaming said he **ate what ate what** at the buffet.'

(35) Syntactic reduplication (Travis 2001, 2003)



Evidence #1

- Deriving the properties:

- (36)
- Phrases only → RED c-selects phrases as complements.
 - Constituency → reduplicaiton occurs at syntx.
 - *Tripllication → there is only one specifier.
 - Plurality → RED is a plural operator.

Evidence #2

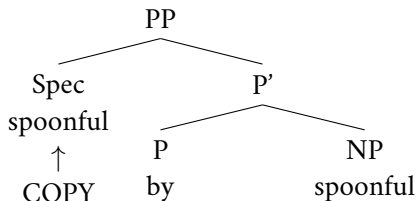
- Parallels with Noun-Preposition-Noun (NPN) constructions in English

(37) NPN constructions in English

- I drank my milk [_{PP} **spoonful** by **spoonful**] . (Huddleston and Pullum 2002, p.633)
- John ate the cake [_{PP} **piece** by **piece**] . (Beck and Stechow 2006, p.45)

- The P in NPN constructions trigger (syntactic) reduplication.

(38) NPN constructions in the spirit of Travis (2001, 2003)



Evidence #2

① The ban on triplication

- (39) a. ***page** for **page** for **page** (Jackendoff 2008, p.21)
- b. *a **baby**-to-**child**-to-**adult** transition (p.21)

Evidence #2

② Tolerating imperfect copy

(40) Imperfect copying in NPN constructions

- a. **step** by **dogged step** (Huddleston and Pullum 2002, p.634)
- b. **day** by/after **miserable day** (Jackendoff 2008, p.21)

(41) Imperfect copying in non-bare *wh*-doublets

阿明話佢嚟自助餐 食咗龍蝦食咗乜野。

Aaming waa keoi hai zizocaan sik-zo lunghaa sik-zo matje .

Aaming say he at buffet eat-PFV lobster eat-PFV what

(lit.): 'Aaming said he at the buffet **ate lobster ate what.**'

Evidence #2

③ Order-sensitivity

(42) Imperfect copying is order-sensitive

a. ***awful day** by/after **day**

(Jackendoff 2008, p.21)

b. *食咗乜野食咗龍蝦

* sik-zo matje sik-zo lunghaa

eat-PFV what eat-PFV lobster

(lit.): ‘ate what ate lobster’

A remark

- Note that this is yet another difference with bare *wh*-doublets.

(43) No imperfect copying in bare *wh*-doublets

*阿明話佢嚟自助餐食咗 龍蝦乜野 。

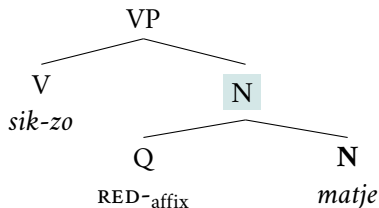
*Aaming waa keoi hai zizocaan sik-zo lunghaa matje

Aaming say he at buffet eat-PFV lobster what

(lit.): 'Aaming said he ate **lobster what** at the buffet.'

Interim summary

(44) Phonological reduplication in bare *wh*-doublets



(45) Syntactic reduplication in non-bare *wh*-doublets

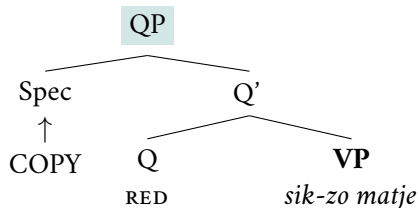


Table of contents

- ① Introduction
- ② Background: *Wh*-doublets
- ③ Bare vs. non-bare
- ④ Plurality
- ⑤ Derivation
- ⑥ Evidential indefinites**
- ⑦ Conclusions

Evidential indefinites

- The indefinite family of *wh*-doublets are known for their subtle differences in meaning.
 - specific indefinites
 - non-specific indefinites
 - free choice items
 - universal quantifiers,
 - free relatives,
 - metalinguistic quantifiers, ...
- In view of recent discussions on indefinites with not-at-issue meaning, it is important to pin down the precise nature of *wh*-doublets in Cantonese.
 - indefinites encoding indifference/ignorance (Kratzer and Shimoyama 2002)
 - epistemic indefinites (Alonso-Ovalle and Menéndez-Benito 2010, 2015)
 - quotative indefinites (Koev 2017)
 - metalinguistic quantifiers (Sudo 2008, 2013)

Evidential indefinites

- We argue that Cantonese *wh*-doublets instantiate a new type of indefinites, which come with an not-at-issue, **evidential implicature**.

→ Evidential indefinites

- We propose to characterize the evidential component as follows.

(46) The evidential component of *wh*-doublets

The use of a *wh*-doublet implies a **previous discourse context** such that the speaker has **access to the source of evidence** about the referent denoted by the *wh*-expression.

- Elaborations on
 - 1 “access to the source of evidence”
 - 2 “a previous discourse context”

#1 Access to the source of evidence

- The idea of “access to the source of evidence” captures why Cantonese *wh*-doublets are most common in quotative/reportative contexts.

(47) 阿明 話 [佢喺自助餐食咗 乜嘢乜嘢] 。

Aaming waa [keoi hai zizocaan sik-zo matje matje]

Aaming say he at buffet ate what what

(lit.): ‘Aaming said he ate **what what** at the buffet.’

- It explains why *wh*-doublets are typically embedded under verbs of saying.

#1 Access to the source of evidence

- But the source of evidence is not restricted to speech/verbal contexts.
- **Non-verbal quotative contexts** also license the use of *wh*-doublets.

① Written source

(48) 報紙 寫住 尋日 邊個邊個 囚犯逃獄。 (Wong 2018, (85))

bouzi se-zyu kamjat bin-go bin-go caufaan
 newspaper write-DUR yesterday which-CL which-CL prisoner
 toujuk.

escape.from.prison

(lit.): 'It is written in the newspaper that **which which** prisoner escaped from the prison.'

#1 Access to the source of evidence

② Visual source - signing

(49) 手語老師 示範 隻手要 點樣點樣 郁先打到手語。

saujyu lousi sifaan zek sau jiu dimjoeng dimjeong

sign.language teacher demonstrate CL hand need how how

juk sin daa-dou-saujyu

move then sign-ACP

(lit.): ‘The sign language teacher demonstrated **how how** move the hand to sign.’

#1 Access to the source of evidence

③ Visual source - video recording

- (50) 閉路電視 影住 個賊向 邊個邊個 方向逃跑。 (Wong 2018, (86))
 bailoudinsi jing-zyu go caak hoeng bin-go bin-go
 CCTV capture-DUR CL thief towards which-CL which-CL
 fonghoeng toupaaau.
 direction run.away
 (lit.): 'It has been captured by the CCTV that the thief runs away towards
which which direction.'

#1 Access to the source of evidence

- Also cases where the source of evidence is unspecified/ merely implied

(51) a. 減肥最重要唔係限制食乜嘢乜嘢。

gaamfei zeoi zungjiu mhai haanzai sik matje matje

diet most important not.be limit eat what what

(lit.): ‘The most important thing about dieting isn’t to limit **what what** to eat.’

b. 生仔之前要注意乜嘢乜嘢，已經令好多人好大壓力。

saangzai zicin jiu zyuzi matje matje, jiging ling

give.birth before need pay.attention what what already cause

hou do jan hou daai aatlik

many people very big stress

(lit.): ‘The consideration of **what what** before giving birth has already stressed out many people.’

#2 Discourse dependence

- The use of *wh*-doublets presupposes a previous discourse context, and thus they are incompatible with out-of-the-blue contexts.
- This property differentiates them from ordinary specific indefinites.

(52) Story beginning

好耐好耐以前，阿明去咗 #邊度邊度 / 某個地方。

hounoi-hounoi-jicin, Aaming heoi-zo #bindou bindou / mau-go deifong

once.upon.a.time, Aaming go-PFV where where/ some CL place

(lit.): 'Once upon a time, Aaming went to #**where where/somewhere**.'

#2 Discourse dependence

- Other out-of-the-blue contexts

(53) Self-introduction

大家好，我係嚟自洛杉機 #邊間邊間 大學／ 某一間 大學嘅阿明。

daaigaahou, ngo hai laizi Loksaamgei #bin-gaan bin-gaan

hello, I be from Los Angeles which CL which CL

daaihok/ mau jat gaan daaihok ge Aaming.

university/ some one CL university MOD Aaming

(lit.): 'Hello. I'm Aaming from #**which which** university in Los Angeles / **some** university in Los Angeles.'

#2 Discourse dependence

- Other out-of-the-blue contexts

(54) Quiz show

世界杯決賽有 #邊隊邊隊 / 有 某一隊 南美洲球隊。係邊一隊？

Saigaaibui kyutcoi jau #bin-deoi bin-deoi / mau-deoi
World.Cup final have which-team which-team/ some-team

Naammeizau kaudeoi. Hai bin-deoi?

South America team. Which one

(lit.): 'The World Cup final has #**which-team which-team** / has **some**
South American team. Which team is it?'

Evidential indefinites

- We thus propose that the not-at-issue component associated with Cantonese *wh*-doublets are slightly different from similar cases in Japanese, Bulgarian and German (Sudo 2008, 2013; Cieschinger and Ebert 2011; Koev 2017).
- The component is best formulated with an evidential character, realized as a conventional implicature in the sense of Potts (2005).

(55) The evidential component of *wh*-doublets

The use of a *wh*-doublet implies a previous *discourse* context such that the speaker has *access to the source of evidence* about the referent denoted by the *wh*-expression.

Table of contents

- 1 Introduction
- 2 Background: *Wh*-doublets
- 3 Bare vs. non-bare
- 4 Plurality
- 5 Derivation
- 6 Evidential indefinites
- 7 Conclusions

Conclusions

- Outlook of *wh*-indefinites in Chinese:

Non-Q- <i>wh</i>	<i>Wh</i> -singlets	Bare doublings	Non-bare doublings
Form	<i>wh</i>	<i>wh-wh</i>	<i>wh+α wh+α</i>
Meaning	number-neutral	number-neutral	plural only
Presupposition	No $\exists$ -entailment	Access to source	Access to source

Conclusions

- A non-uniform derivation, due to different nature of the reduplication morpheme
 - Bare *wh*-doublets → phonological reduplication
 - Non-bare *wh*-doublets → syntactic reduplication

Two RED-morphemes	RED _{-affix}	RED
Reduplication type	Phonological	Syntactic
Morphology	Affix	Free morpheme
Syntax (C-selection)	a <i>wh</i> -expression	a phrase containing a <i>wh</i>
At-issue meaning	An identity function	A plural operator

Table 5: The two proposed RED-morphemes

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